INCH-POUND
MIL-C-14385D(AR)
9 December 1991
SUPERSEDING
MIL-C-14385C(AR)
27 JUNE 1986

MILITARY SPECIFICATION

CANNON, 40MM DUAL AUTOMATIC GUN: M2A1 AND CLIP, CHARGER: M3

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification supports the acquisition of the recoil-operated Cannon, Dual, 40MM: M2A1, consisting of twin barrels with flash suppressors, breech rings, casings, recoil cylinders and loaders, for single and automatic firing and the M3 charger-clip for feeding the cartridges. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SS, Watervliet, N.Y. 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1010

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SPECIFICATIONS

MILITARY

MIL-C-13931 - Cannon, General Specification for

MIL-P-14232 - Parts, Equipment and Tools for Army Material, Packaging of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

MIL-STD-2073-1 - DOD Material Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications.
The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.7)

U.S. Army Armament Research, Development and Engineering Center(ARDEC)

5583768 - Clip, Charger, M3

EPL 5583768 - Engineering Parts List for Clip, Charger, M3

7305371 - Gun, Automatic 40MM M2A1, General Arrangement

EPL 7305371 - Engineering Parts List for Cannon, 40MM Dual Automatic Gun: M2A1

PDS P7305371 - Packaging Data Sheet, Gun, Automatic 40MM, M2A1

Gage 5276650 - Gage, Firing Pin Protrusion

Gage 7247039 - Gage, Headspace

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Gage 7247040 - Gage, Headspace

PUBLICATIONS

Master LQAP 5583768 - Master List of Quality Assurance Provisions for Clip, Charger, M3

Master LQAP 7305371 - Master List of Quality Assurance Provisions for Cannon, 40MM Dual Automatic Gun: M2A1

QAP-APPENDIX -WVA - General Quality Assurance Provisions

(Copies of drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society for Testing and Materials (ASTM)

ASTM D 3951 - Commercial Packaging, Practice for

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, Pennsylvania 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Cannon, clips and component parts shall be in accordance with the applicable drawings 7305371, 5583768, EPL 7305371 and EPL 5583768, this specification, MIL-C-13931, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.3 and 6.2). No first article requirements shall be waived without review and approval by the Procuring

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Contracting Officer (see 6.4).

3.3 Operation.

3.3.1 Cannon. The cannon shall operate within the designed rate of fire without interference, erratic movement or malfunction.

3.3.2 Charger-clips. The assembled clips shall hold each round firmly in proper position. Each hook of the assembled clips shall move to extended position under applied pressure and return to retracted position under spring pressure without lag, sticking or binding.

3.4 Mechanical requirements.

3.4.1 Breech casing bodies. Preparatory to machining the fitted bolt holes, the breech casings shall be so assembled in matched pairs as to insure correct alignment of the trunnions and the recoil guideways.

3.4.2 Firing pins. The point of the firing pins may be finished with crocus cloth but shall not be hand stoned. The protrusion of the firing pin from the face of the breechblock shall be not less than 0.099 inch nor greater than 0.114 inch.

3.4.3 Headspace. Headspace shall be not less than 0.002 inch nor greater than 0.024 inch. (Headspace is the clearance between the forward face of the breechblock and the base of a maximum cartridge when both are in the firing position.)

3.5 Recoil cylinders. The recoil mechanism shall be filled with the type and volume of fluid specified on the recoil cylinder assembly drawing.

3.6 Proof testing. Cannon, replacement tubes, and replacement flash suppressors shall be subject to proof testing.

3.7 Performance.

3.7.1 Firing. Cannon and charger clips shall function properly during single and automatic firing. Irregularities and defects shall be corrected before acceptance.

3.7.2 High pressure resistance. The cannon shall be capable of withstanding a chamber pressure of 48,600 plus or minus 1,700 pounds per square inch when tested as specified in 4.5.2.3 (NOTE: Chamber pressure resistance for automatic firing is 43,000 pounds per square inch).

3.7.3 Firing rate. The firing rate on AUTO-FIRE shall be not less than 120 rounds per minute from each of the cannon, at approximately 70 degrees Fahrenheit ambient temperature.

3.7.4 Recoil. The recoil cylinder of each cannon shall be adjusted so that the recoil shall be not less than 7.4 inches nor more than 8.3 inches.

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3.7.5 Material soundness. After proof firing, all parts shall be free of cracks and fractures and shall comply with the soundness requirements specified on the applicable documents.

3.7.6 Bore enlargement. Bore diameters, measured across the rifled lands after proof firing no more than 15 rounds, shall not exceed the inside (minimum bore) diameter specified on the applicable drawing by more than the enlargement specified in TABLE I.

TABLE I. Bore enlargement.

Distance forward of commencement of rifling	0.10 inch	1 cal.	2 Cal. to muzzle
Enlargement (inch)	0.004	0.004	0.003

NOTE: Interpolate linearly to determine the allowable bore enlargement at distance other than the distances given.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. First Article - Initial Production Test (FA-IPT) (see 4.4).

c. Quality conformance inspection (see 4.5).

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4.3 First article inspection. Unless otherwise specified in the contract, (see 6.2), First Article cannon and specified parts thereof shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with applicable Quality Assurance Provisions (QAPs) and QAP-APPENDIX-WVA. QAPs are listed in Master LQAP 7305371. First article sample sizes shall be as specified in applicable item QAPs, or the contract, as applicable. First article cannon and parts therof shall be representative of the production processes to be used during quantity production.

4.4 First Article - Initial Production Test (FA-IPT). When specified in the contract, (see 6.2), First Article cannon shall be subjected to Initial Production Testing by the Government or any agency specified by the Government and may be conducted at a Government facility to assure compliance with the requirements of this specification and any additional requirements listed in the contract. Initial Production Testing procedures and the sample size of First Article cannon shall be as specified in the contract. First Article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

4.5 Quality conformance inspection.

4.5.1 Examinations.

4.5.1.1 Inspection provisions. Cannon, charger-clips and parts thereof shall be inspected for conformance to requirements of this specification; MIL-C-13931; QAPs listed in Master LQAP 7305371; Master LQAP 5583768; and QAP-APPENDIX-WVA.

4.5.1.2 Breech casing bodies. Alignment of the trunnions and the recoil guideways shall be gaged to assure compliance with 3.4.1.

4.5.1.3 Firing pin protrusion. Protrusion of the firing pin from the face of the breechblock shall be measured with firing pin protrusion gage 5276650 or an approved equivalent, to determine compliance with the requirements of 3.4.2.

4.5.1.4 Headspace. Headspace shall be measured with headspace gages 7247039 and 7247040 or an approved equivalent to determine compliance with the requirement of 3.4.3.

4.5.1.5 Recoil cylinders. The recoil mechanism shall be examined to assure it is filled with the proper type and volume of fluid as specified in 3.5.

4.5.1.6 Final examination. Final examination of each cannon and each charger-clip shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.5.1.7 Inspection of packaging. Inspection of the Packaging requirements of Section 5 shall be performed to assure compliance with Section 5. Non compliance shall be cause for rejection.

4.5.1.8 Inspection approval stamp. The application of the inspection approval

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stamp shall be as specified in MIL-C-13931.

4.5.2 Tests.

4.5.2.1 Functioning of cannons. Each cannon shall be tested to assure smoothness of operation, and absence of interference, binding, erratic movement and malfunction. Failure to comply with the requirements of 3.3.1 shall be cause for rejection.

4.5.2.2 Functioning of charger-clips. Charger-clips shall be tested with dummy rounds to assure compliance with the functional requirements of 3.3.2.

4.5.2.3 Proof testing.

4.5.2.3.1 Cannon. Unless otherwise specified in the contract, each cannon with flash suppressor shall be proof fired by Government personnel at a proving ground designated by TECOM in accordance with TABLE II.

TABLE II. Cannon Firing Schedule.

No. of rounds	Pressure Limit (psi)	Elevation (Deg.)
1	48,600 ± 1,700	0
1	48,600 ± 1,700	75
8 (Auto-Fire)	43,000	75

4.5.2.3.2 Replacement tubes. Unless otherwise specified in the contract, each replacement tube shall be proof fired. The firing schedule shall be in accordance with Table III.

4.5.2.3.3 Replacement flash suppressors. Unless otherwise specified in the contract, each replacement flash suppressor shall be proof fired. The firing schedule shall be in accordance with TABLE III.

Table III. Replacement tube and replacement flash suppressor firing schedule.

No. of Rounds	Pressure Limit (psi)
1	48,600 ± 1,700

As far as practical proof firing of replacement flash suppressors shall be combined with that of replacement tubes. If a combined test is impracticable, the flash suppressors shall be proof fired on a proof facility gun.

4.5.2.3.4 Performance. Cannon and charger clips shall be inspected to assure proper functioning during single and automatic firing to comply with 3.7.1.

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4.5.2.3.5 Firing rate. The firing rate on AUTO-FIRE shall be determined to assure compliance with 3.7.3. The firing rate shall be determined at 75 degrees elevation at approximately 70 degrees Fahrenheit ambient temperature.

4.5.2.3.6 Recoil. The recoil shall be measured to determine compliance with 3.7.4.

4.5.2.3.7 Material soundness. After proof firing, the flash suppressor, breech ring, breechblock, and the exterior surface of the tube shall be magnetic particle tested in accordance with applicable drawings and specifications. In addition, the bore shall be examined with a borescope.

4.5.2.3.8 Bore enlargement. The bore diameters across the lands of the cannon tubes shall be measured after proof firing no more than 15 rounds. Measurement shall be made with a star gage, an air gage, or other approved instrument to determine compliance with 3.7.6 and TABLE I.

4.5.2.3.9 Records. Cannon, replacement tubes, and replacement flash suppressors that pass the firing tests shall be furnished with applicable data on DA Form 2408-4 by the proving ground.

4.5.3 Inspection equipment.

4.5.3.1 Acquisition, maintenance, and disposition. Unless otherwise specified, responsibility for acquisition, maintenance, and disposition of inspection equipment prescribed in applicable item QAPs listed in Master LQAP 7305371 and Master LQAP 5583768, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

5. PACKAGING

5.1 Cannon - Levels A, B and C. Cleaning, drying, preservation, unit packaging and packing of cannon shall be in accordance with the requirements of Packaging Data Sheet (PDS) P7305371 and MIL-P-14232 for the level of protection specified in the contract (see 6.2).

5.2 Repair parts - Level A, B and C. Cleaning, drying, preservation, unit packaging and packing of repair parts, shall be in accordance with the applicable repair part PDS and MIL-P-14232 or the applicable repair part SPI and MIL-STD-2073-1 for the level of protection specified in the contract (see 6.2). When Level C requirements are not shown on the PDS/SPI, Level C packaging shall be in accordance with MIL-STD-1190.

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5.2.1 Level X - Industrial packaging. Cleaning, drying, preservation, unit packaging and packing of repair parts shall be in accordance with ASTM D 3951 when industrial packaging is specified in the contract (see 6.2).

5.2.1.1 Quantity in unit package (QUP). When Level C packaging in accordance with MIL-STD-1190 or industrial packaging of repair parts is specified in the contract, the QUP shall be the same as specified in the contract (see 6.2).

5.3 Marking of packages.

5.3.1 Levels A, B and C. Marking (including bar code marking) of the unit, package, intermediate packages, and exterior shipping containers shall be in accordance with MIL-STD-129.

5.3.2 Level X - industrial. Marking of the unit package, intermediate packages, and exterior shipping containers shall be in accordance with ASTM D 3951 (see 6.2). Bar code marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cannon specified herein is intended for use with the Gun, Antiaircraft, Self-Propelled, Twin, 40MM, M42A1 System.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.2).

d. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

e. All inspection and test data that is required.

f. At whose expense a retest may be performed.

g. Any additional or extended examinations and tests beyond the scope of this specification.

h. Serial numbers for the cannon and repair tube assemblies.

i. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.3).

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j. Availability of inspection equipment from the Government.

k. Extent of contractor's responsibility for Government furnished and for contractor-required final inspection equipment.

l. Applicable acceptance test procedures.

m. When a proof firing sampling plan is authorized, the proof firing sampling plan parameters: initial consecutive quantity (i), and sampling frequency (f) (see 4.5.2.3).

n. Under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

o. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and repair parts (see 5.2).

p. Marking and bar code marking requirements (see 5.3).

q. When warranted, the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

r. Unless otherwise specified, the application of MIL-I-45607 and MIL-STD-45662.

6.3 Consideration of Data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-T-4902	First Article Inspection Report	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the Procuring Contracting Officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot of production items, a standard production item from the contractor's current inventory, and the number of items to be tested. The Procuring Contracting Officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first

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articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Product Assurance Directorate at Watervliet Arsenal.

6.6 Subject term (key word) listing.

Cannon, 40MM: M2A1

Clip, Charger: M3

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading "U.S. Army Armament Research, Development and Engineering Center" (ARDEC) may also include drawings prepared by, and identified as Watervliet Arsenal, Ordnance Corps, Weapons Command, or U.S. Army Armament Research and Development Command, U.S. Army Armament, Munitions and Chemical Command, etc. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army-AR

Preparing activity:

Army-AR

(Project 1010 -A138)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-14385D (AR)	2. DOCUMENT DATE (YYMMDD) 9 DECEMBER 1991

3. DOCUMENT TITLE
CANNON, 40MM DUAL AUTOMATIC GUN: M2A1 AND CLIP, CHARGER; M3

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) ATTN: SMCAR-CCB-SS WATERVLIET, NY 12189-4050	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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